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FACTORY TEST REPORT (出 厂 试 验 报 告)

Supplier name(供应商): Hainan Jinpan Electric Co., Ltd. (海南金盘电气有限公司)

Serial number(生产订单号): 1210014096

Outline drawing number(外形图号): 3T012503

Equipment type(设备型号): Cast coil dry-type transformer(浇注干式变压器)

ZSCB9-2000/6

Rated power(额定容量) (AN): 2000 kVA

Rated voltage(额定电压): 6000/660/660 V

Rated current(额定电流): 192.5/874.8/874.8 A

Rated frequency(额定频率): 50 Hz

Phases(相数): 3

Vector group(联结组别): Dd0yn11

Taps(分接): $\pm 2\%5\%$

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Hainan Jinpan Electric Co., Ltd. (海南金盘电气有限公司)

7 WORKSHOP NO.168 NANHAI AV. HAIKOU, HAINAN, CHINA 570216(中国海南省海口市南海大道168号海口保税区7号厂房)

Compiled by(编制)	杨伟挺	Checked by(校核)	王跃冲	Approved by(批准)	卢毅
Date(日期)	Jul.20,2010	Date(日期)	Jul.20,2010	Date(日期)	Jul.20,2010

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Applicable standards and codes(应用标准和代号):					
✓	IEC 60076-11:2004	Dry-Type Transformers			
	IEEE Std C57.12.91-2001	Standard Test Code for Dry-Type Distribution and Power Transformers			
✓	GB1094.11-2007	Dry-Type Transformers			

Test equipments(试验设备):					
Item (项目)	Description (描述)	Model / Type (型号)	Serial number (序号)	Date last calibrated (上次日期)	Date next calibrated (下次日期)
1	Micro-Ohm meter(微欧计)	WR50-12	292-192	Nov.18, 2009	Nov.17, 2010
2	Ratio bridge(变比电桥)	TR-SPY Mark II	284-231	Nov.18, 2009	Nov.17, 2010
3	High-voltage meter(数字高压表)	SGB-100A	99022	Jul.25, 2009	Jul.24, 2010
4	Digital power meter(功率计)	GWD305D	15583	Jul.25, 2009	Jul.24, 2010
5	PD measurement system(局放仪)	JFD-2B	20050802	Jul.25, 2009	Jul.24, 2011
6	High voltage insulation tester(兆欧表)	3125	1477545	Nov.18, 2009	Nov.17, 2010

1. Winding resistance measurements(绕组电阻测量)					
1.1 Primary windings(一次侧绕组)			Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐		
Tapping(分接)	Resistance (电阻)(Ω)			Maximum imbalance (最大不平衡率)	Ambient temperature (环境温度)
	A - B	B - C	C - A		
A	0.1764	0.1766	0.1763	0.17 %	32.5 °C
B	0.1689	0.1691	0.1688	0.18 %	
C	0.1615	0.1617	0.1614	0.19 %	
D	0.1538	0.1540	0.1537	0.20 %	Humidity (湿度)
E	0.1456	0.1458	0.1455	0.21 %	
57 %					
1.2 Secondary windings(二次侧绕组)			Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐		
Resistance(电阻)(Ω)			Maximum imbalance (最大不平衡率)	Ambient temperature (环境温度)	Humidity (湿度)
a1 - b1	b1 - c1	c1 - a1			
0.001832	0.001830	0.001852	1.20 %	32.5 °C	57 %
Resistance(电阻)(Ω)			Maximum imbalance (最大不平衡率)	Ambient temperature (环境温度)	Humidity (湿度)
a2 - b2	b2 - c2	c2 - a2			
0.002056	0.002056	0.002068	0.58 %	32.5 °C	57 %
Accept criteria (接受标准)		Line-line resistance maximum imbalance ≤ 2% or phase-phase resistance maximum imbalance ≤ 4%.(线电阻最大不平衡率小于2%, 或者相电阻之间的不平衡率小于4%。)			

2. Ratio test(电压比测量)						
			Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐			
Vector group (联结组别)	Tapping (分接)	Primary side (一次侧)(V)	Secondary side (二次侧)(V)	Tolerance (误差) (%)		
				AB/ab	BC/bc	CA/ca
Dyn11	A	6600	660	-0.22	-0.24	-0.23
	B	6300		-0.14	-0.16	-0.15
	C	6000		-0.06	-0.08	-0.07
	D	5700		0.04	0.01	0.03
	E	5400		-0.18	-0.19	-0.19

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Vector group (联结组别)	Tapping (分接)	Primary side (一次侧)(V)	Secondary side (二次侧)(V)	Tolerance (误差) (%)		
				AB/ab	BC/bc	CA/ca
Dd0	A	6600	660	0.06	0.05	0.05
	B	6300		0.13	0.13	0.12
	C	6000		0.21	0.21	0.21
	D	5700		0.30	0.30	0.30
	E	5400		0.08	0.08	0.09

Accept criteria (接受标准)	All taps tolerance within $\pm 0.5\%$. (所有分接容许偏差在 $\pm 0.5\%$ 以内。)
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3. Polarity and phase-relation tests(极性和相位) Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐

Polarity (极性)	Phase-relation (相位)
Subtractive (负极性)	Dd0yn11

Accept criteria (接受标准)	Polarity be showed is subtractive.If LV is Y connection,phase-relation showed is <u>Dyn11</u> If LV is Δ connection,phase-relation showed is <u>Dd0</u> 极性为负极性,当低压Y接时,联结组别为 <u>Dyn11</u> ,当低压为 Δ 接时,联结组别为 <u>Dd0</u> 。
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4. Separate-source AC withstand voltage test (Applied voltage test) (外施耐压试验)

4.1 Primary winding-Secondary winding & earth
(一次绕组 - 二次绕组及地) Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐

Test voltage(试验电压)	Duration(持续时间)	Ambient temperature(环境温度)	Humidity(湿度)
25 kV	60 s	32.5 °C	57 %

4.2 Secondary winding - Primary winding & earth
(二次绕组 - 一次绕组及地) Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐

Test voltage(试验电压)	Duration(持续时间)	Ambient temperature(环境温度)	Humidity(湿度)
5 kV	60 s	32.5 °C	57 %

Accept criteria (接受标准)	The test is successful if no collapse of the test voltage and no change in the measuring current occurs.(试验过程中变压器没有出现闪络和电流突变。)
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5. Induced AC withstand voltage test (Induce voltage test) (感应耐压试验) Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐

Supply from (施压侧)	Test voltage (试验电压)	Frequency (试验频率)	Duration (持续时间)	Ambient temperature (环境温度)	Humidity (湿度)
Sec.side(一次侧)	12000 V	196 Hz	31 s	32.5 °C	57 %

Accept criteria (接受标准)	No abrupt current change under test. (电流没有出现突变。)
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6. No-load loss and excitation current (空载损耗和空载电流测量)

100% rated voltage (100%的额定电压) Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐

No-load loss (空载损耗)	2978 W	No-load current (空载电流)	0.27 %	Ambient temperature (环境温度)	32.5 °C
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Accept criteria (接受标准)	Measured no-load loss and no-load current are in accordance with Spec. (空载损耗和空载电流依据技术协议。)
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7. Load loss and impedance (负载损耗和阻抗电压测量)

Load loss and impedance calculation
(负载损耗和阻抗电压计算)

Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐

Reference temperature (参考温度)	120.0 °C	Load loss (负载损耗)	17090 W	Impedance (阻抗电压)	5.73 %
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Accept criteria (接受标准)	Measured Pf is in accordance with Spec. at rated tap. Impedance is are in accordance with Spec. (负载损耗和阻抗电压依据技术协议。)
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8. Partial discharge measurements(局部放电测量)

Partial discharge measurements of primary side (一次侧局放)

Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐

Ambient temperature (环境温度): 32.5 °C Humidity (湿度): 57 %

Frequency (频率)	Applied side (施压侧)	Duration (持续时间)	Inception voltage (激发电压)	A Phase (相)	B Phase (相)	C Phase (相)
196 Hz	Primary side (一次侧)	30 s	1.8UN(V) 10800	4 pC	3 pC	3 pC
		Duration (持续时间)	Extinction voltage (熄灭电压)	A Phase (相)	B Phase (相)	C Phase (相)
		180 s	1.3UN(V) 7800	4 pC	3 pC	3 pC

Accept criteria (接受标准)	The PD value measured is less than 5 pC. (局部放电量小于5pC。)
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9. Insulation resistance measurements (绝缘电阻测量)

Pass(合格) ☒ Fail(不合格) ☐ N/A(不要求) ☐

Ambient temperature (环境温度): 32.5 °C Humidity (湿度): 57 %

Windings (绕组)	Insulation resistance (绝缘电阻)
Primary side - Secondary side and Earth (一次侧对二次侧及地)	100000 MΩ
Secondary side - Primary side and Earth (二次侧对一次侧及地)	100000 MΩ
Primary side and Secondary side - Earth (一次侧和二次侧对地)	100000 MΩ
Core - Earth (铁心对地)	516 MΩ

Accept criteria (接受标准)	Primary windings ≥3000 MΩ, Secondary windings ≥600 MΩ, Core ≥2 MΩ. (一次绕组 ≥3000 MΩ, 二次绕组 ≥600 MΩ, 铁心 ≥2 MΩ。)
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Summary: All factory tests have been passed successfully according to the specifications issued by customer, This unit is qualified and ready for delivery.

IEC 60076-11:2004 GB1094.11-2007

本产品经试验符合标准及合同要求, 准予出厂。

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